系统建模与仿真 第 1 次作业

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用两种仿真思想计算 pi 的值

浦丰的针

平行线间距：a，针的长度：l，投掷次数：n，相交次数：m

```
a = 10;
l = 5;
n = 22200
```

```
n =
22200
```

生成随机位置：distance 表示针中心与平行线的距离，angle 表示针与平行线所成角度（均为均匀分布）

```
distance = rand(1,n)*a/2
```

```
distance = 1×22200
    4.6291    0.8536    0.9843    2.1401    1.0832    4.1550    1.0887    1.4901 ···
```

```
angle = rand(1,n)*90
```

```
angle = 1×22200
   49.0766   88.4088   51.4380   47.8785   15.8541   28.4176   36.5855   61.0580 ···
```

判断与平行线相交的针的个数

```
judge = distance<(l/2).*sind(angle)
```

```
judge = 1×22200 logical 数□
   0   1   1   0   0   0   1   1   1   0   1   0   1   0   1   0   0   1   0   0 ···
```

```
m = length(judge(judge==1))
```

```
m =
7030
```

从理论推导公式，用频率估计概率，算出 pi

```
mypi = 2*l*n/m/a
```

```
mypi =
3.1579
```

向正方形内随机投点

随机生成横纵坐标均位于[-1,1]的若干点

```
number=50540
```

```
number =
50540
```

```
x = rand(1,number)*2-1
```

```
x = 1×50540
   -0.0972    0.3236   -0.0549    0.5617    0.0844    0.9027    0.2165    0.1672 ···
```

```
y = rand(1,number)*2-1
```

```
y = 1×50540
    0.2278   -0.5500   -0.5399   -0.7219   -0.4945   -0.2404    0.6127   -0.7234 ···
```

判断点是否在圆内，记录个数

```
isinround = sqrt(x.^2 + y.^2) < 1
```

```
isinround = 1×50540 logical 数□
   1   1   1   1   1   1   1   1   1   1   0   1   1   1   1   1   1   0   1   1 ···
```

```
numberinround = length(isinround(isinround==1))
```

```
numberinround =
39728
```

计算 pi

```
mysecondpi = numberinround/number*4
```

```
mysecondpi =
3.1443
```